PREDICTIVE DATA SCIENCE

CASE STUDY AND PRACTICAL ASSIGNMENT

P.S.R. ENGINEERING COLLEGE, SIVAKASI

Department of Computer Science and Engineering

Course Code	231CSV65
Course Name	Predictive Data Science
Academic Year	2026–2027
Case Study	Predicting Patient Re-admission in Healthcare

1. Case Study

a) Identify the problem faced by the hospital. (2 Marks)

Johns Hopkins Hospital faces frequent patient readmissions within 30 days of discharge. These readmissions can increase healthcare costs, put additional pressure on hospital resources, and indicate that some patients may need more support after leaving the hospital. The hospital therefore wants to identify patients at higher risk so preventive care can be arranged.

b) Explain the input data used for building the predictive model. (3 Marks)

The case study mentions EHR data including patient medical history, laboratory results, and hospital stay details. Medical history describes previous illnesses and chronic conditions; laboratory results provide measurements about the patient's condition; and hospital stay details describe the current admission and discharge. Data should be checked for errors and missing values and handled securely.

c) Describe how predictive analytics helps identify high-risk patients. (4 Marks)

Predictive analytics uses patterns in historical records to estimate future readmission risk. The workflow includes preparing EHR data, selecting features, training a model on past records, evaluating it on separate data, and applying it to a newly discharged patient. The output is a risk estimate that can flag patients for follow-up. It is not a guarantee and does not replace clinical judgment.

d) Discuss the actions taken and the benefits of using predictive analytics in healthcare. (5 Marks)

For higher-risk patients, providers may arrange earlier follow-up, explain discharge instructions, review medication plans, coordinate care, and make follow-up calls. Benefits can include earlier intervention, improved continuity of care, better use of staff time and resources, and a possible reduction in avoidable readmissions and costs. Performance, fairness and privacy should be monitored.

e) Interpret the chart and explain the role of predictive analytics in reducing patient readmissions. (5 Marks)

The supplied college PDF refers to a chart, but the chart is not present in the supplied PDF. Therefore, the original chart cannot be interpreted from the provided material. In a typical illustrative chart, a lower readmission rate after risk-based follow-up may suggest that targeted support helps; a higher rate in a high-risk group may show why risk scores can prioritize support. A chart alone does not prove causation.

f) Analyze how the predicted risk scores help healthcare providers make decisions. (5 Marks)

A risk score summarizes the model's estimate of a patient's likelihood of readmission. Providers may use it to prioritize calls, schedule earlier reviews, coordinate services and identify patients needing extra discharge support. Thresholds should reflect resources and the consequences of missing a high-risk patient. Scores are estimates and should be combined with clinical judgment and checks for error or bias.

2. Practical Programs

• Simple Linear Regression — Python

• Logistic Regression — Python

• Decision Tree — Python

• SVM — Python

• Naive Bayes — Python

• Market Basket Analysis / Association Rules — Python

• Breast Cancer Prediction — R

• HR Analytics — R

3. Verified Python Outputs

These logs were generated by executing the included Python programs. Re-run locally for submission screenshots if required.

simple_linear_regression.py

```
================================================
SIMPLE LINEAR REGRESSION
================================================
Number of records : 8
Regression equation: Score = 26.79 + 7.38 × Hours
R-squared         : 0.995
Predicted score for 6.5 study hours: 74.76
Note: Illustrative dataset for learning only.

Process exit code: 0
```

logistic_regression.py

```
================================================
LOGISTIC REGRESSION
================================================
Number of records : 10
Training accuracy : 100.0%
Predicted pass probability (6 hours, 88% attendance): 1.000
Predicted class    : PASS
Note: Illustrative dataset; training accuracy is not a test-set score.

Process exit code: 0
```

decision_tree_classifier.py

```
================================================
DECISION TREE CLASSIFIER
================================================
Chosen split       : attendance <= 67.50
Training accuracy  : 100.0%
Prediction (5 study hours, 82% attendance): PASS
Note: Deliberately small, one-split teaching tree.

Process exit code: 0
```

svm_classifier.py

```
================================================
LINEAR SVM CLASSIFIER
================================================
Number of records : 10
Training accuracy : 80.0%
Prediction (5 study hours, 82% attendance): PASS
Note: Simple hinge-loss gradient descent teaching implementation.

Process exit code: 0
```

naive_bayes.py

```
================================================
GAUSSIAN NAIVE BAYES
================================================
Number of records : 10
Training accuracy : 100.0%
Prediction (5 study hours, 82% attendance): PASS
Note: Illustrative dataset; training accuracy is not a test-set score.
Process exit code: 0
```

market_basket_analysis.py

```
============================================================
MARKET BASKET ANALYSIS — ASSOCIATION RULES
============================================================
Number of transactions: 5
Minimum support: 0.40
Minimum confidence: 0.60
Frequent itemset ['beer']: support = 0.60
Frequent itemset ['bread']: support = 0.80
Frequent itemset ['cola']: support = 0.40
Frequent itemset ['diaper']: support = 0.80
Frequent itemset ['milk']: support = 0.80
Frequent itemset ['beer', 'bread']: support = 0.40
Frequent itemset ['beer', 'diaper']: support = 0.60
Frequent itemset ['beer', 'milk']: support = 0.40
Frequent itemset ['bread', 'diaper']: support = 0.60
Frequent itemset ['bread', 'milk']: support = 0.60
Frequent itemset ['cola', 'diaper']: support = 0.40
Frequent itemset ['cola', 'milk']: support = 0.40
Frequent itemset ['diaper', 'milk']: support = 0.60
Frequent itemset ['beer', 'bread', 'diaper']: support = 0.40
Frequent itemset ['beer', 'diaper', 'milk']: support = 0.40
Frequent itemset ['bread', 'diaper', 'milk']: support = 0.40
Frequent itemset ['cola', 'diaper', 'milk']: support = 0.40

Association rules:
['beer'] -> ['bread'] | confidence = 0.67, lift = 0.83
['beer'] -> ['diaper'] | confidence = 1.00, lift = 1.25
['diaper'] -> ['beer'] | confidence = 0.75, lift = 1.25
['beer'] -> ['milk'] | confidence = 0.67, lift = 0.83
['bread'] -> ['diaper'] | confidence = 0.75, lift = 0.94
['diaper'] -> ['bread'] | confidence = 0.75, lift = 0.94
['bread'] -> ['milk'] | confidence = 0.75, lift = 0.94
['milk'] -> ['bread'] | confidence = 0.75, lift = 0.94
['cola'] -> ['diaper'] | confidence = 1.00, lift = 1.25
['cola'] -> ['milk'] | confidence = 1.00, lift = 1.25
['diaper'] -> ['milk'] | confidence = 0.75, lift = 0.94
['milk'] -> ['diaper'] | confidence = 0.75, lift = 0.94
['beer'] -> ['bread', 'diaper'] | confidence = 0.67, lift = 1.11
['beer', 'bread'] -> ['diaper'] | confidence = 1.00, lift = 1.25
['beer', 'diaper'] -> ['bread'] | confidence = 0.67, lift = 0.83
['bread', 'diaper'] -> ['beer'] | confidence = 0.67, lift = 1.11
['beer'] -> ['diaper', 'milk'] | confidence = 0.67, lift = 1.11
['beer', 'diaper'] -> ['milk'] | confidence = 0.67, lift = 0.83
['beer', 'milk'] -> ['diaper'] | confidence = 1.00, lift = 1.25
['diaper', 'milk'] -> ['beer'] | confidence = 0.67, lift = 1.11
['bread', 'diaper'] -> ['milk'] | confidence = 0.67, lift = 0.83
['bread', 'milk'] -> ['diaper'] | confidence = 0.67, lift = 0.83
['diaper', 'milk'] -> ['bread'] | confidence = 0.67, lift = 0.83
['cola'] -> ['diaper', 'milk'] | confidence = 1.00, lift = 1.67
['cola', 'diaper'] -> ['milk'] | confidence = 1.00, lift = 1.25
['cola', 'milk'] -> ['diaper'] | confidence = 1.00, lift = 1.25
['diaper', 'milk'] -> ['cola'] | confidence = 0.67, lift = 1.67
Interpretation: lift > 1 indicates positive association in this sample.
Note: Tiny illustrative dataset, not a business recommendation.

Process exit code: 0
```

4. Limitations and Submission Notes

• Datasets are small illustrative teaching data, not real patient or business data.

• Reported classification accuracy is training-set performance on a tiny sample, not a reliable estimate of unseen-data performance.

• The breast-cancer program is not a diagnostic system.

• The HR trend is a classroom forecast, not a real staffing recommendation.

• The chart mentioned in question (e) was not present in the supplied PDF, so no fabricated chart interpretation is provided.

• Run the Python programs in VS Code and the R programs in RStudio before taking your final local screenshots.